2024 27th International Conference on Computer and Information Technology (ICCIT)
20-22 December 2024, Cox's Bazar, Bangladesh

2024 27th International Conference on Computer and Information Technology (ICCIT) | 979-8-3315-1909-4/24/$31.00 ©2024 IEEE | DOI: 10.1109/ICCIT64611.2024.11021790

Personalized Teaching Strategy Suggestion for Toddlers with ASD: A Machine Learning and Deep Learning Approach

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***Abstract*—Autism Spectrum Disorder (ASD) is a neuro-developmental disorders which have impacts particularly on social interaction, communication, cognitive development, and adaptive learning. Since every infant with ASD has different autism screening characteristics and skill gaps, it is necessary to choose the best teaching strategy in order to meet their educational demands. In order to fulfill the developmental needs of each toddler with ASD, we want to determine the best teaching strategies that are well suited according to their learning requirements. To provide a suitable teaching strategy for toddlers, this study introduces an approach of Machine Learning (ML) and Deep Learning (DL) which analyzes the distinctive characteristics of autism. We have used here Analysis of Variance (ANOVA) feature selection technique to see how each feature influences the target variable. We have applied Decision Tree (DT), Random Forest Classifier (RFC), Feedforward Neural Network (FFNN), Deep Neural Network (DNN) for classifying the most appropriate teaching strategy. The FFNN and DNN performs outstanding with 100% accuracy.**

Keywords: Autism Spectrum Disorder (ASD), Teaching Strategy, Autism screening characteristics, Machine Leaning (ML), Deep Learning (DL), Decision Tree (DT) , Random Forest Classifier (RFC), Feedforward Neural Network (FFNN), Deep Neural Network (DNN).

I. Introduction

Autism Spectrum Disorder (ASD) is a developmental disability that often manifests in early childhood and can persist throughout a person's life. ASD can be characterized by restricted and repetitive activities [1]. According to UNESCO, education is the basic human right of every child [2]. Besides focusing on detection of ASD in children, we must also ensure the educational rights for children in autism syndrome. Since each child with ASD needs a customized education, it is important to identify an appropriate teaching strategy for these children. Even Paulette Delgado explains in her article that schools cannot fulfill the need of special children who need special treatment [3]. In schools, social interactions are common and always evolving. Furthermore, some activities in the classroom could be appropriate for students in general but inappropriate for autistic child. However, an autistic child typically finds it challenging to comply with specific social cues or indicators that ask them to alter their particular actions. Due to a lack of comprehensive knowledge, educators are confused about how to handle an autistic child who requires special education services. This could have a detrimental effect on their schooling [4].

It is crucial to identify better teaching-strategies to support the learning of children with ASD, as many of them struggle in mainstream classrooms [4]. If appropriate teaching strategies are followed for their learning, then this will effectively contribute to the success of ASD children's lives. There are different teaching strategies for children with ASD. These teaching strategies will help them learn things effectively as well as improve their social engagement. We will focus on suggesting effective teaching strategy for ASD children based on age and autism screening characteristics, as each individual toddler have unique needs and characteristics. To explore more extensive and effective identification and give suggestions of teaching approaches for ASD children, we explore DT, RFC, FFNN, DNN models. Identifying the specific educational needs of children with ASD can be challenging and often lacks precision using traditional methods. However, machine learning algorithms offer a more efficient solution by processing large sets of variables with greater accuracy. These algorithms are capable of analyzing complex data patterns and identifying tailored educational strategies that meet the unique needs of each child more precisely than conventional approaches.

II. Background Study and Literature Review

This section will focus on the previous relevant works on Autism children's education. Vasileios, Joseph and Xiao [5] had researched on the topic of 1 "effective teacher communication strategies" slection in the education for autism children and discovered that majority of contact between teachers and students in a classroom typically takes place verbally [6]. Researchers in an East London special school for children with ASD observed 5,460 instances of teacher-student interactions, analyzing teaching goals, strategies, emotional states, and reactions. Using machine learning models like logistic regression and random forest, they predicted student responses based on

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teaching styles and communication methods. However, the study focused primarily on verbal communication, which is often ineffective for children with ASD, and did not address individual skill deficiencies or tailored teaching strategies [5]. In this paper author had examined teacher communication strategies and used machine learning to predict student responses based on communication style and emotional state. However, the study focused mainly on verbal communication, which may be ineffective for many children with ASD, and did not adequately address individual developmental skill deficiencies or how to tailor teaching strategies to those needs.

The study [7] proposes a novel framework for the early Autism detection and the selection of appropriate teaching methods for children with ASD , using multiple ML classifiers, such as K-Nearest Neighbor (KNN), Random Forest (RF), Logistic Regression (LR), Decision Tree (DT), Support Vector Machine (SVM), Gradient Boosting Machine (GBM), XGBoost (XGB), and a combined RF-XG model, and a varied dataset created for toddlers. To handle class imbalance, they applied SMOTE, and feature selection methods like Boruta and ML-based Feature Selection (MFS). The RF-XG model achieved high accuracy (0.9917) in recommending teaching methods based on autism severity [7].

This study proposed a machine learning framework for early autism detection and recommending appropriate teaching methods based on a child's autism severity. Although the study achieved high accuracy in identifying teaching methods, it did not explicitly address the unique developmental challenges that toddlers face, particularly in terms of mapping teaching strategies to specific skill deficiencies.

In the paper [4] authors studied how machine learning can be used to identify the best teaching methods for children with ASD. Their goal was to predict the most effective teaching strategies for each child based on extensive data. They used algorithms like K-Nearest Neighbor, Decision Tree, Naïve Bayes, and Random Forest to evaluate six teaching techniques. The Random Forest algorithm performed the best, showing that machine learning can effectively tailor educational interventions. Their findings highlight the potential for personalized, inclusive teaching approaches that can improve the educational outcomes for children with ASD. [4]. In this paper there is no clear explanation of recommending appropriate education strategies according to the autism characteristics.

The study [8] provides valuable insights into the use of AI for enhancing the education of children with autism, focusing on collaboration between educators and therapists, and AI. However it, lacks a detailed exploration of how machine learning models can deliver personalized teaching strategies based on individual needs, and it does not fully address practical implementation challenges, such as integration of diverse datasets. These limitations present opportunities for further research, which our study aims to address by focusing on personalized, data-driven strategies.

A common limitation among these studies is that they didn't focus on the specific developmental skill deficiencies of toddlers with ASD. As a result, teaching strategies may not fully address the child's individual needs. In order to address this gap, this study directly matches teaching methods with the developmental skills that need the greatest attention, making sure that the strategies selected are highly relevant to the unique requirements of toddlers. The goal of this strategy is to offer a more successful and individualized educational experience at the stage of developmental age.

From the preceding section it is clear that there are huge opportunity to contribute in this area by employing ML and DL based models for identification of suitable education methods for children with ASD. Majority of the work outlined above has some limitations.

III. Proposed Methodology

In this section, we will describe the proposed methodology for the prediction of education strategy for autism toddlers from the dataset based on the autism screening characteristics and relevant education mapping. The screening data set contains data of age groups of 12 months to 36 months. Our methodology is depicted in a flowchart which is shown below in Fig 1. The flowchart outlines the phases of methodology resulting in the research workflow in sequence.

A. Data Collection and Preparation

1) Data collection: The proposed methodology being started by implementing with collection of datasets for toddlers with ASD.We use here 2 datasets that will span the age group from 12 to 36 months and aspects of autism screening and behavior analysis: "Behavior Analyis of Autism" [9], "Toddler Autism dataset" [10]. Those datasets are created based on Q-CHAT10 for toddles, where the questions are validated in the paper [11].

2) Data Preparation: We have prepared a dataset where the mapping the skills with it's relevant teaching approaches.

- Toddlers_Edu_mapped_to_skill.csv

In this dataset, the attributes of each Autism Screening in Q-CHAT10(for Toddlers) are mapped to the skills where it is denoting the deficiency in certain skills. Figure 2 shows the teaching strategies and their corresponding skills that are usually enhanced by them. There is a column that shows score of Teaching strategis of covering skills. We have created a new column in each dataset for ASD toddlers [9]- [10] named **"Teaching strategy"** based on created mapping csv dataset. The column will contain only that has the education-strategy highest score of enhancing targeted skills.

The table I provides an overview of our data preparation procedure, mapping skill deficiencies to relevant teaching strategies and scoring them based on the skills they address.

B. Data Preprocessing

To ensure most accurate result the dataset should be clean and standard.For this reason data preprocessing is crucial. We have conducted Label encoding for the categorical features, dropped the unnecessary columns so to reduce the complexity of model and trained with the most effective and correlated features.

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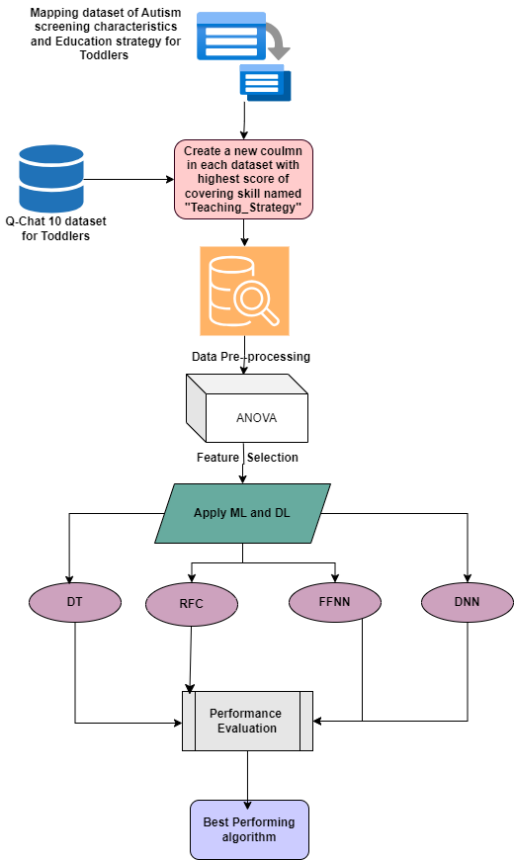


Fig. 1. The proposed methodology flowchart.



Fig. 2. Teaching Strategies and their corresponding enhancing skills

TABLE I
SCREENING TRAITS OF TODDLERS(Q-CHAT10), CHALLENGES AND TEACHING STRATEGY SCORES

Screen characteristic	Characteristic problems outlined	Deficient skills	Ref.	Strategy score
A1	Infant with ASD have problem in responding when calling their names	responding to others	[12]	PMII=1
A2	For many persons with autism, making eye-contact can be very difficult	eye contact	[13]	PECS=1
A3	This is imperative pointing, The lackings of imperative pointing indicates the problem "joint_attention"	joint attention	[14]	TA=1, TAII=1, PECS=1
A4	The shared focus used to express interest in an object is known as the declarative function of joint attention, lack of declarative pointing indicates the problem "joint_attention"	joint attention	[14]	TA=1, TAII=1, PECS=1
A5	Pretending is associated with play skills and adaptive functioning skills	play skill, adaptive	[15]	PRT=1, TA=1, TAII=1
A6	The ability to listen and follow directions involves the cognitive ability called executive function. Cognitive impairments are common in ASD	cognitive	[16]	TAII=1
A7	The persons with autism usually are faces difficulties in communication, understanding, and interacting with others, which can lead to a lack of empathy	communication, understanding others, interacting with others	[17]	PRT=1, TAII=1, PECS=1, PMII=2
A8	There is a problem with language and communication skill	language, communication	[18]	PRT, PECS=2; TA, TAII=1
A9	The gesture is associated with social and adaptive behavior	social, adaptive	[19]	TA=2, TAII=2, PECS=1
A10	The toddlers with ASD commonly "space out" and begin to stare blankly. This is caused by sensory processing issues.	sensory integration	[20]	PRT=1

C. Feature Selection

For finding the features that are strongly related with the target variable by comparing their F-values. Higher F-values indicates that the feature is more relevant in predicting the target. According to the ANOVA results for Toddler and Behavior dataset, we have selected our most prominent features.

From the figure [3] and [4], we can figure out the followings:
• Top Features: Top features for both datasets are mostly similar, such as A6, A8, A2, and Qchat-10-Score are the most significant. But blood group also appears significant in behavior dataset.
• P-values:These features are also statistically significant (p is less than 0.05), showing their strong influence on teaching strategies. We took them for combining ANOVA results for the new combined dataset.

Combined ranking of features for both datasets

Figure [5] shows the ranking of features like toddler datasets. The top-ranked features, such as A6 (executive function), A8 (language/communication), A2 (eye contact), and Qchat-10-Score, are highly significant in predicting teaching strategies because they directly relate to key areas of de-

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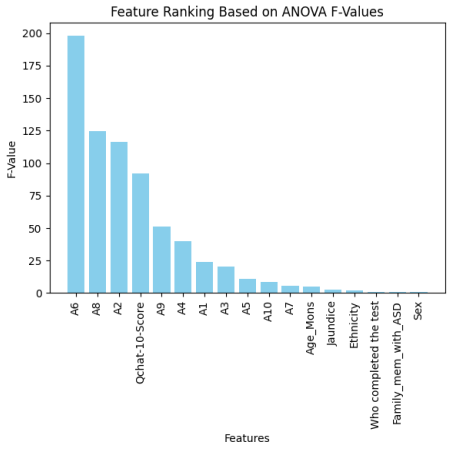


Fig. 3. Feature ranking results for Toddler dataset using ANOVA

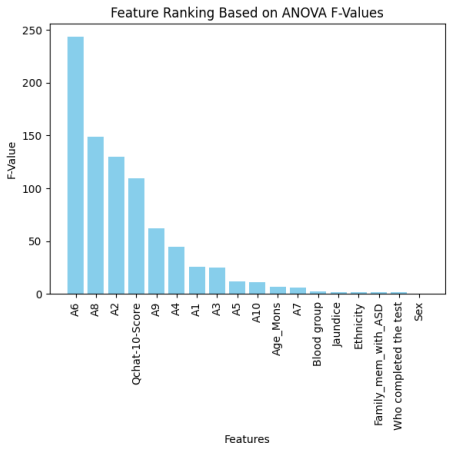


Fig. 4. Feature ranking results for Behavior dataset using ANOVA

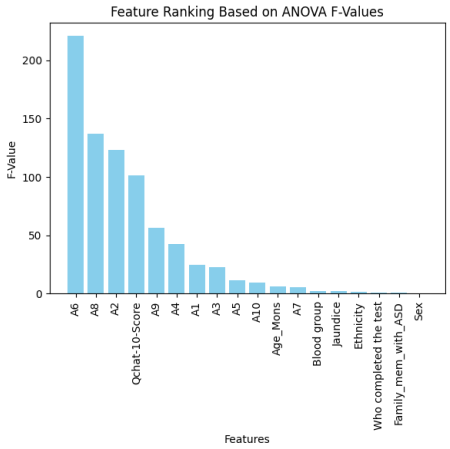


Fig. 5. Combined feature ranking results for both dataset using ANOVA

velopment in children with autism. These features influence essential skills like following instructions, communication, and social interactions, which are critical to designing effective personalized teaching approaches.

In our implementation part we have all the common features for both dataset and also have taken first 13 ranked features in count. The result for 13 ranked is performed more perfectly than all the features. Then we merged the two datasets having 2242 instances and 14 columns (top ranked 12 features, Case_no, and newly created target variable column teaching_strategy) , We have split them into 70% and 30% respectively for training our models and testing the prediction performance of the models.

IV. Result

The results for taking all features without the ANOVA techniques produce the result: that shows in the table II.

TABLE II
Results of ML and DL models for all features

Models	Accuracy%	Prescision%	Recall%	F1-score%
DT	**99.14**	**99.16**	**99.14**	**99.14**
RFC	98.71	98.75	98.71	98.62
FFNN	98.07	98.11	98.07	97.94
DNN	98.71	98.73	98.71	98.71

The table III shows the performances of ML and DL models after using the ANOVA results for feature selection.

TABLE III
Results of ML and DL models for 13 top most features selected by ANOVA

Models	Accuracy%	Prescision%	Recall%	F1-score%
DT	99.14	99.16	99.14	99.14
RFC	99.14	99.16	99.14	99.10
FFNN	**100**	**100**	**100**	**100**
DNN	**100**	**100**	**100**	**100**

The DT is consistent in both cases of using features but the neural network models are outperformed in the selected feature scenario. The feature selection impacts the performances of the models mostly in FFNN and DNN. By selecting the most relevant features like A6 (executive function), A8 (language/communication), and Qchat-10-Score , we can reduce the noise and improve interpretability. These features address critical skill deficiencies in children with ASD, aligning with effective teaching strategies. Focusing on significant features improved the models' accuracy and interpretability, ensuring reliable, personalized recommendations. The reason is the neural network model's ability to capture complex relationships, which enhances its practical value, providing educators with clear and trustworthy insights for real-world application. Here in Table IV depicts the State-of-the-art (SOAT) comparisons for the studies that we are suggesting teaching strategies for ASD child based on ML and DL applications.

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TABLE IV
COMPARISON OF PROPOSED MODEL WITH STATE-OF-THE-ART (SOTA) METHODS

Author	Dataset	Exploring-dataset	Result
Fahima Hajjej [7]	1.Toddler Autism dataset [10] 2. ASD screening data for toddlers in Saudi Arabia	Yes	DT: 96.88% RFC:98.44% FFNN:98.44% DNN:99.07%
Zarin Tassnim Zoana [4]	1.Toddler Autism dataset [10] 2.Behavior Analysis of Autism [9] 3.Child data 2017 and 18 [21]	Yes (partially) 3rd dataset is not align with our focus age range	DT:98.69 (reported in paper)% RFC:98.69 (reported in paper)%
Our approach	1.Toddler Autism dataset [10] 2.Behavior Analysis of Autism [9]	–	DT:99.14% RFC:99.14% FFNN:100% DNN:100%

V. CONCLUSION AND FUTURE WORKS

This study demonstrates the effectiveness of machine learning and deep learning models in developing personalized teaching strategies for toddlers with ASD. By using ANOVA for feature selection, we were able to enhance the accuracy of the models, particularly for deep learning approaches, which achieved 100% accuracy in predicting tailored teaching strategies. The ANOVA test identified the most statistically significant features those that have the strongest relationship with the target variable (teaching strategies). Since these features contribute the most to the prediction, removing the less significant ones allows the model to focus on what matters, improving performance. The models successfully mapped autism screening characteristics to teaching strategies that address the skill deficiencies of each child, providing a powerful tool for educators and parents. Our study aims to identify tailored teaching strategies for toddlers, utilizing the QChat10 datasets that are available on Kaggle. While the datasets are valuable, their limited size may not encompass the full spectrum of ASD variations. We will try to expand the diversity of the datasets, and will collect datasets on Bangladeshi autistic infants' data for further analysis. Our study shows how ML and DL can help to select personalized teaching strategies for toddlers with Autism.

Future Works: While our current work focuses on tabular data, we have plan to expand our research to:

- Use image and video data to analyze the behaviors of ASD infants.
- Use transformer-based models to analyze image and video data for better insights into non-verbal behaviors.
- Expnad our datasets to include more diverse children and autism characteristics.
- Develop a web application for giving user interface to choose personalized education for child with ASD very easily and at early stage.
- Conduct a study in a specialized ASD school using our model to gather feedback, ensuring practical validation and improving its effectiveness.

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